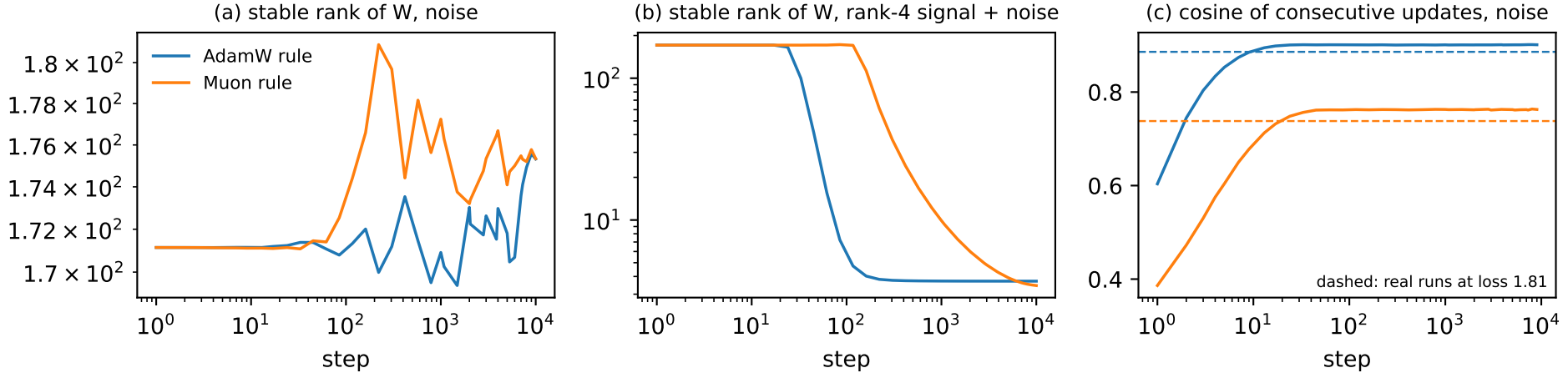
(a) stable rank of W, noise
AdamW rule
Muon rule
1.8×10^2
1.78×10^2
1.76×10^2
1.74×10^2
1.72×10^2
1.7×10^2
10^0 10^1 10^2 10^3 10^4
step
(b) stable rank of W, rank-4 signal + noise
10^2
10^1
10^0 10^1 10^2 10^3 10^4
step
(c) cosine of consecutive updates, noise
0.8
0.6
0.4
dashed: real runs at loss 1.81
10^0 10^1 10^2 10^3 10^4
step